MACHINE LEARNING

(Practical 2)

SOURCE CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('emails.csv')
df.head()

df.isnull().sum()

df.dropna(how='any',inplace=True)

x = df.iloc[:,1:-1].values
y = df.iloc[:,-1].values

from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.25,random
_state=10)

from sklearn.metrics import PrecisionRecallDisplay, RocCurveDisplay

from sklearn.metrics import (
    ConfusionMatrixDisplay,
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    PrecisionRecallDisplay,
    RocCurveDisplay
)

def report(classifier):
    y_pred = classifier.predict(x_test)
    cm = confusion_matrix(y_test, y_pred)

    display = ConfusionMatrixDisplay(cm, display_labels=classifier.classes
_)
    display.plot()

    print(f"Accuracy:         {accuracy_score(y_test, y_pred):.4f}")
```

```
    print(f"Precision Score:  {precision_score(y_test, y_pred):.4f}")
    print(f"Recall Score:     {recall_score(y_test, y_pred):.4f}")

    PrecisionRecallDisplay.from_estimator(classifier, x_test, y_test)
    RocCurveDisplay.from_estimator(classifier, x_test, y_test)

from sklearn.neighbors import KNeighborsClassifier

kNN = KNeighborsClassifier(n_neighbors=10)
kNN.fit(x_train,y_train)

report(kNN)

from sklearn.svm import SVC
svm = SVC(gamma='auto',random_state=10)
svm.fit(x_train,y_train)

report(svm)
```

OUTPUT

In [1]: df = pd.read_csv('emails.csv'); df.head()

Out [1]:

	Email No.	the	to	ect	and	for	of	a	you	hou	.. .	conne vey	jay	valu ed	lay	infrastruc ture	milita ry	allowi ng	ff	dr y	Predict ion
0	Email 1	0	0	1	0	0	0	2	0	0	.. .	0	0	0	0	0	0	0	0	0	0
1	Email 2	8	13	24	6	6	2	102	1	27	.. .	0	0	0	0	0	0	0	1	0	0
2	Email 3	0	0	1	0	0	0	8	0	0	.. .	0	0	0	0	0	0	0	0	0	0
3	Email 4	0	5	22	0	5	1	51	2	10	.. .	0	0	0	0	0	0	0	0	0	0
4	Email 5	7	6	17	1	5	2	57	0	9	.. .	0	0	0	0	0	0	0	1	0	0

5 rows × 3002 columns

In [2]: df.isnull().sum()

Out [2]:

```
Email No.     0
the           0
to            0
ect           0
and           0
              ..
military      0
allowing      0
ff            0
dry           0
Prediction    0
Length: 3002, dtype: int64
```

In [8]: kNN = KNeighborsClassifier(n_neighbors=10); kNN.fit(x_train, y_train)

Out [8]:

```
KNeighborsClassifier(n_neighbors=10)
```

In [9]: report(kNN)

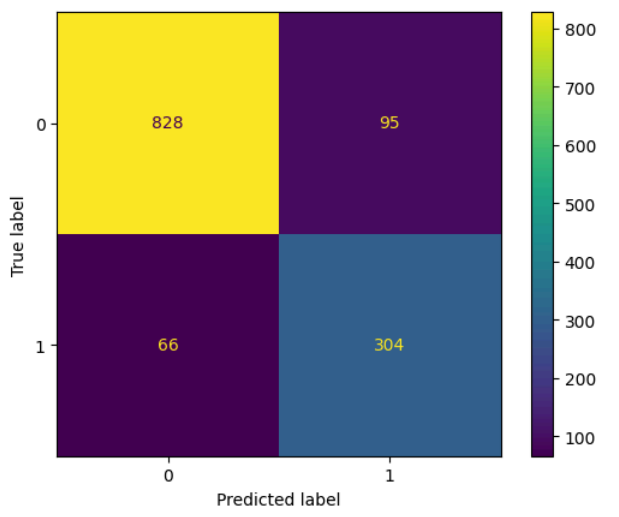
Performance Metrics (K-Nearest Neighbors):

Accuracy: 0.8755

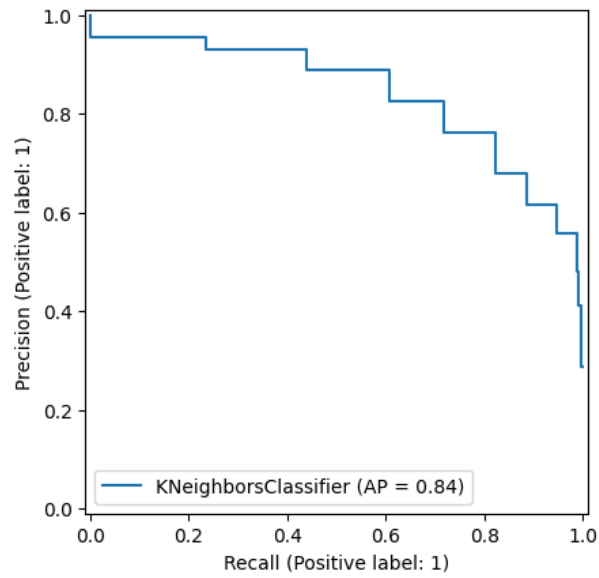
Precision Score: 0.7619

Recall Score: 0.8216

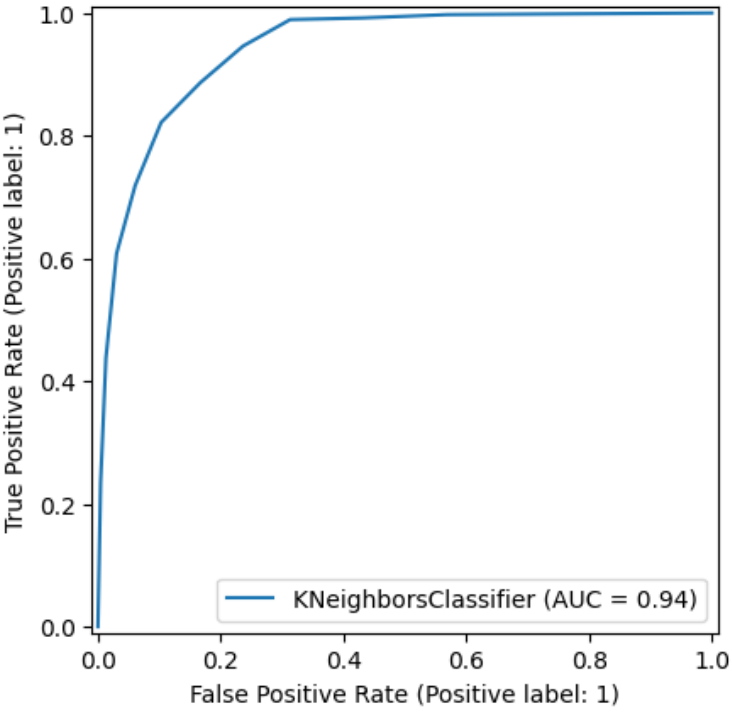
<Figure size 640x480 with 2 Axes><Figure size 640x480 with 1 Axes><Figure size 640x480 with 1 Axes>



Confusion Matrix (kNN)



Precision-Recall Curve (kNN)



ROC Curve (kNN)

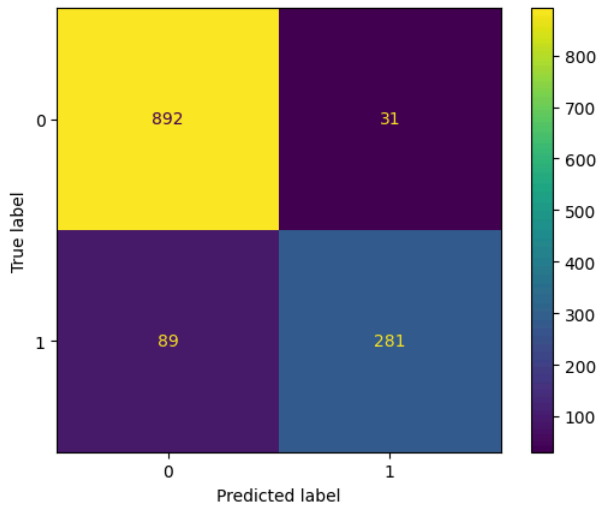
In [10]: from sklearn.svm import SVC; svm = SVC(gamma='auto', random_state=10); svm.fit(x_train, y_train)

Out [10]:

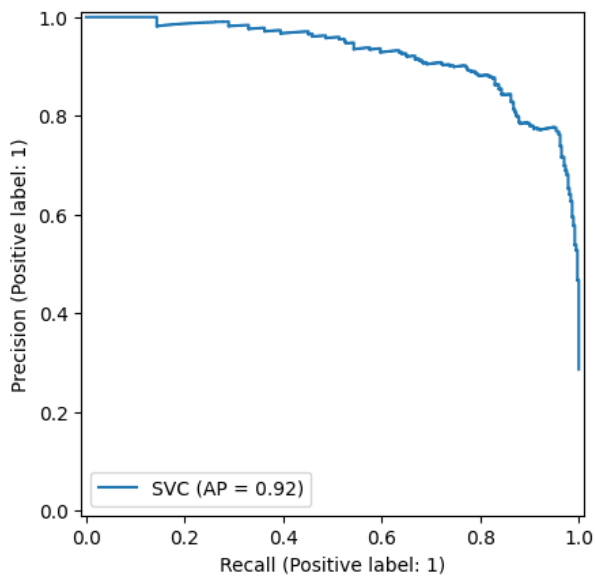
```
SVC(gamma='auto', random_state=10)
```

In [11]: report(svm)

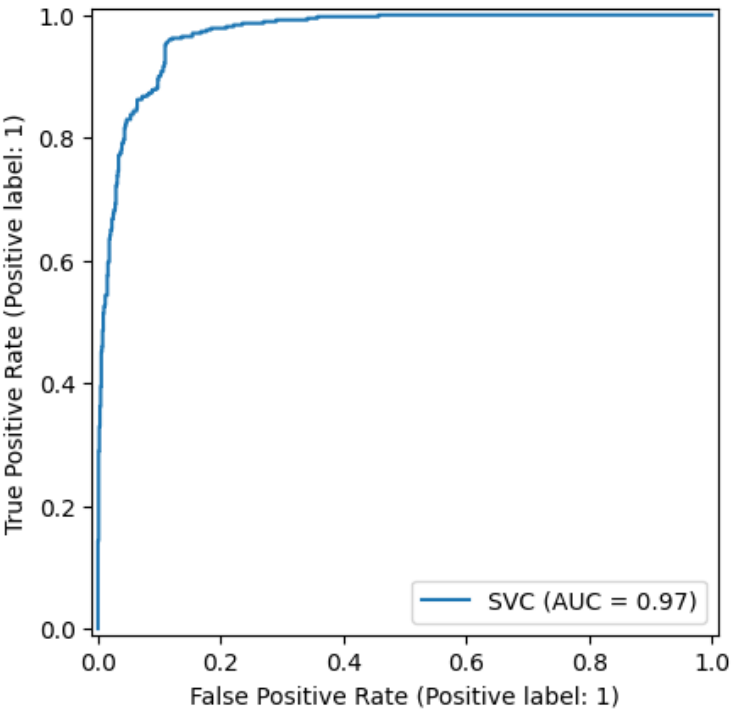
Performance Metrics (Support Vector Machine):
Accuracy: 0.9072
Precision Score: 0.9006
Recall Score: 0.7595
<Figure size 640x480 with 2 Axes><Figure size 640x480 with 1 Axes><Figure size 640x480 with 1 Axes>



Confusion Matrix (SVM)



Precision-Recall Curve (SVM)



ROC Curve (SVM)